MANAGING PROCESSES AND OPTIMIZING PERFORMANCE

WHAT IS PROCESS

A process is a running executable programme. whenever the executable programme is in running state called as process.

TYPES OF PROCESSES

1. Foreground Process
2. Background Process
3. Daemon process

Foreground Process

A foreground process is a process that runs in the terminal and interacts directly with the user.

1. It takes input from the user
2.It displays output on the screen
3.The terminal is busy until the process finishes

Example:
1.sleep 20

Background Process

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A background process is a process that runs without user interaction and does not block the terminal.

1.It runs behind the scenes
2.The terminal remains free for other commands
3.Usually started using &

Example:
1.sleep 50 &

2.jobs

[1]+ Running sleep 50 & ----------> run in background

[1]+ Done sleep 50 ----------> task is done

Daemon Process

1.A daemon process is a background process that runs continuously and provides system or network services without user interaction.
2.These processes generally started at boot time

2.Examples: apache/nginx, sshd, crond

Example on Background and Foreground

Run task on foreground

1.sleep 200
ctrl+z

To check the jobs

1.jobs
[1]+ Stopped sleep 200

Run stopped job in background

3.bg
[1]+ Running sleep 200 &

Run running job in foreground

4.fg
sleep 200
ctrl+z

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To check the jobs

5.jobs
[1]+ Stopped sleep 200

Run stopped job in background

6.bg
[1]+ Running sleep 200 &

TO RUN A SPECIFIC JOBS

1.sleep 500
ctrl+z
2.sleep 600
ctrl+z
3.sleep 700

ctrl+z

4.jobs

```
[1]   Stopped          sleep 500
[2]- Stopped           sleep 600
[3]+ Stopped           sleep 800
```

5.To run a specific 2nd job in background

bg %2

6.jobs

```
[1]   Stopped          sleep 500
[2]- Running           sleep 600
[3]+  Stopped          sleep 800
```

7.To run a specific 2nd job in foreground

fg %2

TO CHECK THE SYSTEM PROCESSES

1.ps -aux

- Shows all running processes in the system
- Includes processes of all users like ur root user/system user/daemon user
- Displays the CPU, memory, and command related details
- Shows the static output like screenshot
- **Command: ps -aux**
  1. ps → process status
  2. a → all users
  3. u → user format (detailed)
  4. x → includes background processes

2.ps -elf

- Displays processes in very detailed format
- Shows parent-child relationship
- Includes priority and nice values
- Useful for deep analysis
- It also shows the static data like screenshot
- **Command: ps -elf**
  1. ps → process status
  2. e → all processes
  3. l → long format
  4. f → full (tree-like hierarchy)

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3.Top

- Shows live running processes which is refreshable within 2seconds of interval
- Displays CPU and memory usage
- Updates automatically in real-time
- Highlights top resource usage
- **Command Explanation:**
  1. top → table of processes
  2. Press q → quit
  3. Press k → kill process

4.htop

- Advanced version of the top command
- User-friendly and colourful interface
- Supports mouse and keyboard navigation
- Shows process tree and CPU cores
- **Command Explanation:**
  1. htop → interactive process viewer
  2. Needs installation (apt install htop)
  3. Use arrow keys to navigate
  4. Press F10 → exit

STATES OF PROCESSES

1. ***Running process: (R)***
   - Currently using CPU
   - Actively executing instructions

2. ***Stopped process: (T)***
   - Paused by user or system
   - Example: Ctrl + Z
   - Can be resumed anytime

3. ***Sleep process: (S) (interruptible)***
   - Waiting for input or resource
   - Can be interrupted easily
   - Most common state
   - Example: waiting for keyboard input

4. ***Sleep process: (D) (uninterruptable)***
   - Process is waiting for disk or hardware work
   - Cannot be stopped until work is finished
   - System must complete the task first

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5. ***zombie process: (Z)***
    - The process which is killed but still shows it is present
    - Process has finished execution (killed)
    - But still exists in process table
    - Happens when parent process doesn't collect exit status

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COMMAND TO TERMINATE PROCESS

1.kill

It is used to terminate a process manually

Commands:

1.kill <pid> ----------> terminate a process
2.kill -9 <pid> ----------> terminate a process forcefully
3.pkill <process name> ----------> terminate a process-by-process name

Example:

1.sleep 500
2.ps

```
  PID TTY          TIME CMD
  901 pts/2    00:00:00 bash
  920 pts/2    00:00:00 sleep
```

3.kill -9 920
4.ps

```
  PID TTY          TIME CMD
  901 pts/2    00:00:00 bash
```

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Explanation Of (ps -elf) Command

F S UID PID PPID C PRI NI ADDR SZ WCHAN STIME TTY TIME CMD

1. F -------> system internal info
2. S -------> states of processes
3. UID ----> user id (which user started these processes)
4. PID ----> process id
5. PPID ---> parent process id (If bash starts a command → bash is parent)
6. C -------> CPU usage
7. PRI -----> priority to process (Lower value = higher priority)
8. NI -------> nice value (controls the priority) (-20 to 19)

9. ADDR ------> Memory address
10. SZ ----------> Memory size (Memory size used by process (in pages)
11. WCHAN --> Wait channel (What the process is waiting for)
12. STIME ----> Start Time (When the process is start)
13. TTY --------> Terminal type
    - pts/0 → SSH session
    - tty1 → physical terminal
    - ? → no terminal (background process)
14. TIME --------> Total CPU time consumed
15. CMD ---------> Command (Process name)
